Experiment 9

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Aim

To create and implement PL/SQL packages by developing a package specification and package body containing procedures and shared cursors, in order to achieve modular, reusable, and efficient database programming.

Software Requirements

- Database Management System:
  - Oracle Database Express Edition (Oracle XE)
  - PostgreSQL Database
- Database Administration Tool / Client Tool:
  - Oracle SQL Developer (for Oracle XE)
  - pgAdmin (for PostgreSQL)

Objectives

To design and implement a PL/SQL package that includes procedures and shared cursors for structured and modular program development.

Procedure

1. **Initialization of Environment**
   The Oracle database environment was started using SQL Developer. A connection was established with the required database to perform the experiment.
2. **Creation of Employee Table**
   A table named *employee* was created with appropriate attributes such as

employee ID, name, salary, and department. The employee ID was defined as the primary key to uniquely identify each record.

3. **Insertion of Sample Data**
   Sample records were inserted into the employee table to facilitate testing of the package procedures and cursor operations.
4. **Creation of Package Specification**
   A package specification was designed to declare the procedures that can be accessed externally. This included procedures for fetching all employee details and retrieving a specific employee based on ID.
5. **Creation of Package Body**
   The package body was implemented to define the actual logic of the procedures declared in the specification. This included the implementation of procedures and database operations.
6. **Implementation of Shared Cursor**
   A shared cursor was created within the package body to retrieve employee records. This cursor was reused by different procedures, ensuring modularity and reducing redundancy.
7. **Implementation of Procedures**
   Procedures were developed inside the package to display all employee records and to fetch details of a specific employee. These procedures utilized the shared cursor and SQL queries.
8. **Execution of Package Procedures**
   The procedures defined in the package were executed using PL/SQL blocks. The output was displayed using standard output mechanisms.
9. **Verification of Results**
   The results obtained from executing the procedures were verified to ensure that the correct employee data was retrieved and displayed.
10. **Testing and Validation**
    Multiple test cases were performed by retrieving different employee records to validate the correctness, efficiency, and reusability of the package.

Input/Output Analysis

SQL Input Queries

CREATE SCHEMA emp_package;

```
Data Output   Messages   Notifications

CREATE SCHEMA

Query returned successfully in 71 msec.
```

CREATE TABLE employee (

```
    emp_id NUMBER PRIMARY KEY,
    emp_name VARCHAR2(50),
    salary NUMBER,
    department VARCHAR2(50)
);
```

Output

```
Data Output   Messages   Notifications

CREATE TABLE

Query returned successfully in 47 msec.
```

INSERT INTO employee VALUES (1, 'Sejal', 50000, 'AI');
INSERT INTO employee VALUES (2, 'Rahul', 45000, 'IT');
INSERT INTO employee VALUES (3, 'Priya', 60000, 'HR');

COMMIT;

Output

```
Data Output   Messages   Notifications

INSERT 0 3

Query returned successfully in 88 msec.
```

```
CREATE OR REPLACE PACKAGE emp_package AS

    -- Procedure to display all employees
    PROCEDURE get_all_employees;

    -- Procedure to display employee by ID
    PROCEDURE get_employee_by_id(p_id NUMBER);

END emp_package;
```

Output

```
Data Output   Messages   Notifications

CREATE FUNCTION

Query returned successfully in 133 msec.
```

```
CREATE OR REPLACE PACKAGE BODY emp_package AS

    -- Shared Cursor
    CURSOR emp_cursor IS
        SELECT emp_id, emp_name, salary, department FROM employee;

    -- Procedure to display all employees
    PROCEDURE get_all_employees IS
    BEGIN
        FOR rec IN emp_cursor LOOP
            DBMS_OUTPUT.PUT_LINE(
```

```
            'ID: ' || rec.emp_id ||
            ', Name: ' || rec.emp_name ||
            ', Salary: ' || rec.salary ||
            ', Dept: ' || rec.department
        );
    END LOOP;
END;

-- Procedure to display employee by ID
PROCEDURE get_employee_by_id(p_id NUMBER) IS
BEGIN
    FOR rec IN (SELECT * FROM employee WHERE emp_id = p_id) LOOP
        DBMS_OUTPUT.PUT_LINE(
            'ID: ' || rec.emp_id ||
            ', Name: ' || rec.emp_name ||
            ', Salary: ' || rec.salary ||
            ', Dept: ' || rec.department
        );
    END LOOP;
END;

END emp_package;
/
```

Output

Data Output Messages Notifications

```
CREATE FUNCTION

Query returned successfully in 75 msec.
```

```
SET SERVEROUTPUT ON;
```

Output

```
Data Output   Messages   Notifications

CREATE PROCEDURE

Query returned successfully in 38 msec.
```

```
-- DISPLAY ALL EMPLOYEES
BEGIN
   EMP_PACKAGE.GET_ALL_EMPLOYEES;
END;
/

-- DISPLAY SPECIFIC EMPLOYEE
BEGIN
   EMP_PACKAGE.GET_EMPLOYEE_BY_ID(1);
END;
/
```

```
Data Output   Messages   Notifications

COMMIT

Query returned successfully in 78 msec.
```

Learning Outcomes

- Understood PL/SQL package structure

- Learned difference between specification and body
- Implemented shared cursor
- Developed modular and reusable code
- Gained industry-relevant skills